

Complex Variables Practice Problem Set

Sections 8 & 10 (Brown & Churchill)

Section 8: Arguments of Products and Quotients

1. Find the principal argument $\text{Arg } z$ when

(a) $z = \frac{i}{-2 - 2i}$;

(b) $z = (\sqrt{3} - i)^6$.

3. Use mathematical induction to show that

$$e^{i\theta_1} e^{i\theta_2} \dots e^{i\theta_n} = e^{i(\theta_1 + \theta_2 + \dots + \theta_n)} \quad (n = 2, 3, \dots).$$

4. Using the fact that the modulus $|e^{i\theta} - 1|$ is the distance between the points $e^{i\theta}$ and 1 (see Sec. 4), give a geometric argument to find a value of θ in the interval $0 \leq \theta < 2\pi$ that satisfies the equation $|e^{i\theta} - 1| = 2$.

6. Show that if $\text{Re } z_1 > 0$ and $\text{Re } z_2 > 0$, then

$$\text{Arg}(z_1 z_2) = \text{Arg } z_1 + \text{Arg } z_2,$$

where principal arguments are used.

9. Establish the identity

$$1 + z + z^2 + \dots + z^n = \frac{1 - z^{n+1}}{1 - z} \quad (z \neq 1)$$

and then use it to derive *Lagrange's trigonometric identity*:

$$1 + \cos \theta + \cos 2\theta + \dots + \cos n\theta = \frac{1}{2} + \frac{\sin[(2n+1)\theta/2]}{2 \sin(\theta/2)} \quad (0 < \theta < 2\pi).$$

10. Use de Moivre's formula (Sec. 7) to derive the following trigonometric identities:

(a) $\cos 3\theta = \cos^3 \theta - 3 \cos \theta \sin^2 \theta$;

(b) $\sin 3\theta = 3 \cos^2 \theta \sin \theta - \sin^3 \theta$.

Section 10: Roots of Complex Numbers

2. In each case, find all the roots in rectangular coordinates, exhibit them as vertices of certain squares, and point out which is the principal root:

(a) $(-16)^{1/4}$;

(b) $(-8 - 8\sqrt{3}i)^{1/4}$.

3. In each case, find all the roots in rectangular coordinates, exhibit them as vertices of certain regular polygons, and identify the principal root:

- (a) $(-1)^{1/3}$;
- (b) $8^{1/6}$.

6. Find the four zeros of the polynomial $z^4 + 4$, one of them being

$$z_0 = \sqrt{2}e^{i\pi/4} = 1 + i.$$

Then use those zeros to factor $z^4 + 4$ into quadratic factors with real coefficients.

8. (a) Prove that the usual formula solves the quadratic equation

$$az^2 + bz + c = 0 \quad (a \neq 0)$$

when the coefficients a, b , and c are complex numbers. Specifically, by completing the square on the left-hand side, derive the quadratic formula

$$z = \frac{-b + (b^2 - 4ac)^{1/2}}{2a},$$

where both square roots are to be considered when $b^2 - 4ac \neq 0$.

(b) Use the result in part (a) to find the roots of the equation $z^2 + 2z + (1 - i) = 0$.

A1. If $z \in \mathbf{C}$, $r > 0$ and $n \in \mathbf{N}$, then show that

$$\left(\frac{z}{r^n}\right)^{1/n} = \frac{z^{1/n}}{r}.$$